

Shi Luode

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Research Interests: Multi-Agent Multi-Objective Optimization, Reinforcement Learning, and Autonomous Robotics

EDUCATION

National University of Singapore (NUS)

08/2025–Present

Master of Science in Robotics | GPA: 4.07/5.00 | Expected graduation: Nov 28, 2026

- ◆ Research focus: autonomous exploration, reinforcement learning, decision optimization, viewpoint planning, and graph-based policy learning.
- ◆ Relevant coursework: Topics in Robotics (A+), AI Planning and Decision Making, Cognitive Robotics, Robot Dynamics and Control, and Machine Learning in Robotics.

Shandong University (SDU)

09/2021–06/2025

B.Eng. in Optical Information Science and Engineering | GPA: 3.63/4.00 | Rank: 8/41

RESEARCH EXPERIENCE

PAMAO: Preference-Adaptive Multi-Agent Optimization for Multi-Objective Combinatorial Optimization

04/2026–09/2026

Research Project, MARMoT Lab, NUS; arXiv preprint available, planned submission to ICML 2027

- ◆ Research problem: a shared preference sets the desired balance between objectives, such as shorter total travel versus more balanced routes, but each vehicle or machine has a different role. Developed PAMAO to adapt decisions to each agent and to learn from good solutions found under nearby trade-offs.
- ◆ Method design: added two components to MACSIM and PARCO without replacing their original encoders or action decoders. Agent-wise Preference Injection (AWPI) adjusts each agent's decision using its role; Collaborative Self-Improvement (CSI) compares candidate solutions from nearby preferences under the same target trade-off, then reuses the best joint plan for training.
- ◆ Experimental results: evaluated routing (MOMTSP, MOHCVRP) and flexible job-shop scheduling (MOMAFJSP), measuring solution-set quality and trade-off coverage with hypervolume. On generated MOMTSP, 9 combinations of 60/80/100 customers and 3/5/8 agents (200 test instances each), PAMAO-M improved on MACSIM in 9/9 and surpassed PARCO in 9/9. On six generated two-objective scheduling settings (200 instances each), it improved on MACSIM in 6/6; under DCAN's official greedy evaluation, PAMAO-D improved on DCAN in 6/6 two-objective settings, 5/6 three-objective in-range settings, and 6/6 larger-scale settings.
- ◆ Public benchmarks: evaluated 84 cases (24 routing, 20 vehicle-routing, and 40 scheduling). PAMAO-M improved the average solution-set score across all three benchmark groups; on the 20-case vehicle-routing set, the score rose from 0.8288 to 0.8298 while recorded time fell from 292.2 to 169.6 seconds per instance.

Ariadne2: Reinforcement Learning for Single-Robot Autonomous Exploration

12/2025–05/2026

Research Assistant, MARMoT Laboratory, National University of Singapore

Advisor: Asst. Prof. Guillaume Sartoretti | Robotics, Autonomous Exploration, Reinforcement Learning

- ◆ Goal: enable a robot to explore an unfamiliar map by choosing useful observation points, rather than repeatedly moving to the nearest unexplored boundary.
- ◆ Planning system: generated candidate points around map boundaries, snapped them to traversable locations, removed duplicates, ranked them by newly visible area, and grouped nearby candidates; capped the action list at 25 points per decision.
- ◆ Training method: built an expert planner that selects points with low overlap in newly visible regions and orders them to reduce travel. Combined its routes with exploration-coverage and travel-distance rewards to train a Soft Actor-Critic policy with two value networks.
- ◆ Implementation and evaluation: integrated the map graph, candidate planner, policy, and parallel simulation in Ray. Tested on 100 maps with up to 600 decisions per run and recorded map-completion rate and travel distance.

ADDITIONAL EXPERIENCE

Machine Vision Intern, Shandong Institutes of Industrial Technology

07/2024–08/2024

Industrial inspection and machine vision

- ◆ Built a production-line inspection workflow for open glass bottles, from camera calibration and image threshold setup through defect filtering and result reporting.
- ◆ Used MXVision and image-region size/shape rules to identify empty positions, inverted bottles, and bottles that had not been opened; tuned the checks with production samples.
- ◆ Processed inspection results with C# scripts and delivered bottle counts to the client; also assisted with defect-inspection tasks for Hikrobot products.

TECHNICAL SKILLS

- ◆ **Reinforcement Learning:** decision-process modeling, Soft Actor-Critic, twin value networks, reward design, expert-route imitation, and policies guided by objective trade-offs.
- ◆ **Multi-Agent Optimization:** multi-agent routing and job-shop scheduling; preference-guided neural solvers, agent-role adaptation, collaborative training, and solution-quality evaluation.
- ◆ **Robotics and Exploration:** occupancy-map exploration, frontier and viewpoint selection, graph-based planning, candidate-action generation, route coverage, and A*-based path planning.
- ◆ **Machine Learning Systems:** Python, PyTorch, Transformer models, Ray, Hydra, TensorBoard, Lightning/RL4CO, parallel simulation and training, and unit testing.
- ◆ **Development Tools and English:** Codex and AI coding tools for code reading, experiment scripts, debugging, and technical writing; IELTS 6.5, CET-4, and CET-6.

HONORS AND AWARDS

Optoelectronics Alumni Scholarship, School of Information Science and Engineering, Shandong University

2025

Second-Class Academic Scholarship, Shandong University

2024

Outstanding Participant, SIOM Summer Camp, University of Chinese Academy of Sciences

2024

Second-Class Academic Scholarship, Shandong University

2023

Third-Class Academic Scholarship, Shandong University

2022